SYSDESIGN REFERENCE

Design a Distributed Job Scheduler

Requirements-to-architecture reference walkthrough.

Execute durable jobs at or after a target time with ownership, retries, and visibility.

VERSION 1.0 REVIEW

REVIEWED THROUGH 2026-07-18

STATUS HUMAN REVIEW REQUIRED

SYSDESIGN / JOB-SCHEDULER

SYSDESIGN

Design a Distributed Job Scheduler

CORE IDEA

Execute durable jobs at or after a target time with ownership, retries, and visibility.

Reference architecture decision chain

01	02	03	04
Time Buckets	Leasing	Queues	Idempotency

Decisions to defend

+ Time buckets
+ leasing
+ queues
+ idempotency
+ priorities
+ and backfill.

Failure modes to test

+ Duplicate execution
+ missed jobs
+ clock skew
+ and retry pileups.

Interview flow

- [ ] Clarify users, critical paths, and non-goals
- [ ] Estimate only scale that changes a decision
- [ ] Define API, data model, and source of truth
- [ ] Draw synchronous and asynchronous paths
- [ ] Pressure-test timeout, retry, overload, and recovery
- [ ] Close with tradeoffs and one alternative

Deep-dive drill

Design a Distributed Job Scheduler

Probe	What a strong answer makes explicit
How does the design behave during a regional dependency failure?	Requirement, decision, cost, failure behavior, and observable metric.
Which metric reveals the first bottleneck?	Requirement, decision, cost, failure behavior, and observable metric.
What consistency does the user actually observe?	Requirement, decision, cost, failure behavior, and observable metric.
How would the design change at ten times the traffic?	Requirement, decision, cost, failure behavior, and observable metric.

Data questions

+ Partition key
+ Versioning
+ Uniqueness
+ Retention
+ Reconciliation

Operations questions

+ SLO
+ Backlog
+ Capacity
+ Alert
+ Failover test

DESIGN DISCIPLINE

For every box: which requirement justifies it, what cost does it introduce, and what happens when it fails?

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